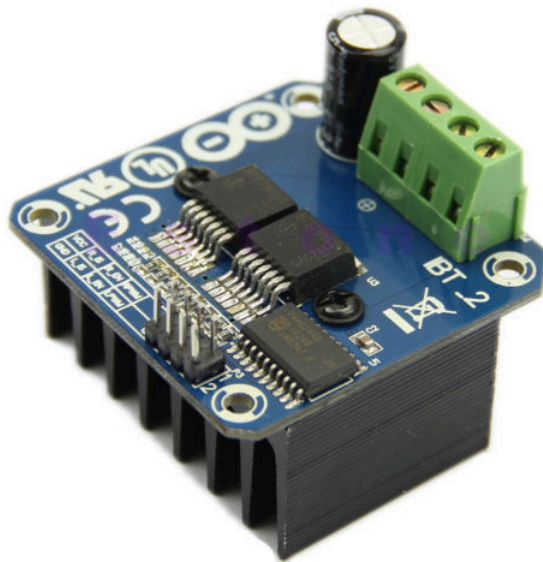


Dr Rainer Hessmer

Robotics, Software, and more

IBT-2 H-Bridge with Arduino

The IBT-2 H-bridge module from wingxin is an inexpensive, high power motor driver based on two BTS7960 chips and is readily available from various ebay vendors; see e.g. [here](#).



The [link](#) provides more details but here are a few key parameters.

- Input voltage : 6V-27V
- Maximum Current : 43A
- Input level : 3.3V-5V

I am not sure whether the heat sink is sufficient for handling 43A but even if one does not drive the unit to its limits the specifications are still impressive given the typical price point of this module (currently between \$13 and \$17 including free shipping from China). There is relatively little information available about how to hook up the module with an Arduino controller. This [thread on the Arduino forum](#) was useful but the solution wastes a few pins and does not demonstrate bidirectional motor control. In this post I describe a slightly more complete solution

that uses an Arduino controller with connected potentiometer to drive a motor via the IBT-2 module from full reverse speed to full forward speed.

For reference here is the description of the input ports and the two supported usage modes (image taken from [one of the ebay postings](#)). In this post I leverage usage mode one.

Input port

1	2	1、 RPWM	: Forward level or PWM signal input, active high
2	3	2、 LPWM	: Inversion level or PWM signal input, active high
3	4	3、 R_EN	: Forward drive enable input , high enable , low close
4	5	4、 L_EN	: Reverse drive enable input , high enable , low close
5	6	5、 R_IS	: Forward drive –side current alarm output
6	7	6、 L_IS	: Reverse drive –side current alarm output
7	8	7、 VCC	: +5 V power input, connected to the microcontroller 5V power supply
		8、 GND	: Signal common ground terminal

Usage one:

VCC pick MCU 5V power supply, GND connected microcontroller GND
 R_EN and L_EN shorted and connected to 5V level, the drive to work.
 L_PWM, input PWM signal or high motor forward
 R_PWM, input PWM signal or high motor reversal

Usage two:

VCC pick MCU 5V power supply , GND connected microcontroller GND
 R_EN and L_EN short circuit and PWM signal input connected to high-speed
 L_PWM, pin input 5V level motor is transferred
 R_PWM, pin input 5V level motor reversal

Here is the associated Arduino sketch:

```

1  /*
2  IBT-2 Motor Control Board driven by Arduino.
3
4  Speed and direction controlled by a potentiometer attac
5  One side pin of the potentiometer (either one) to groun
6
7  Connection to the IBT-2 board:
8  IBT-2 pin 1 (RPWM) to Arduino pin 5(PWM)

```

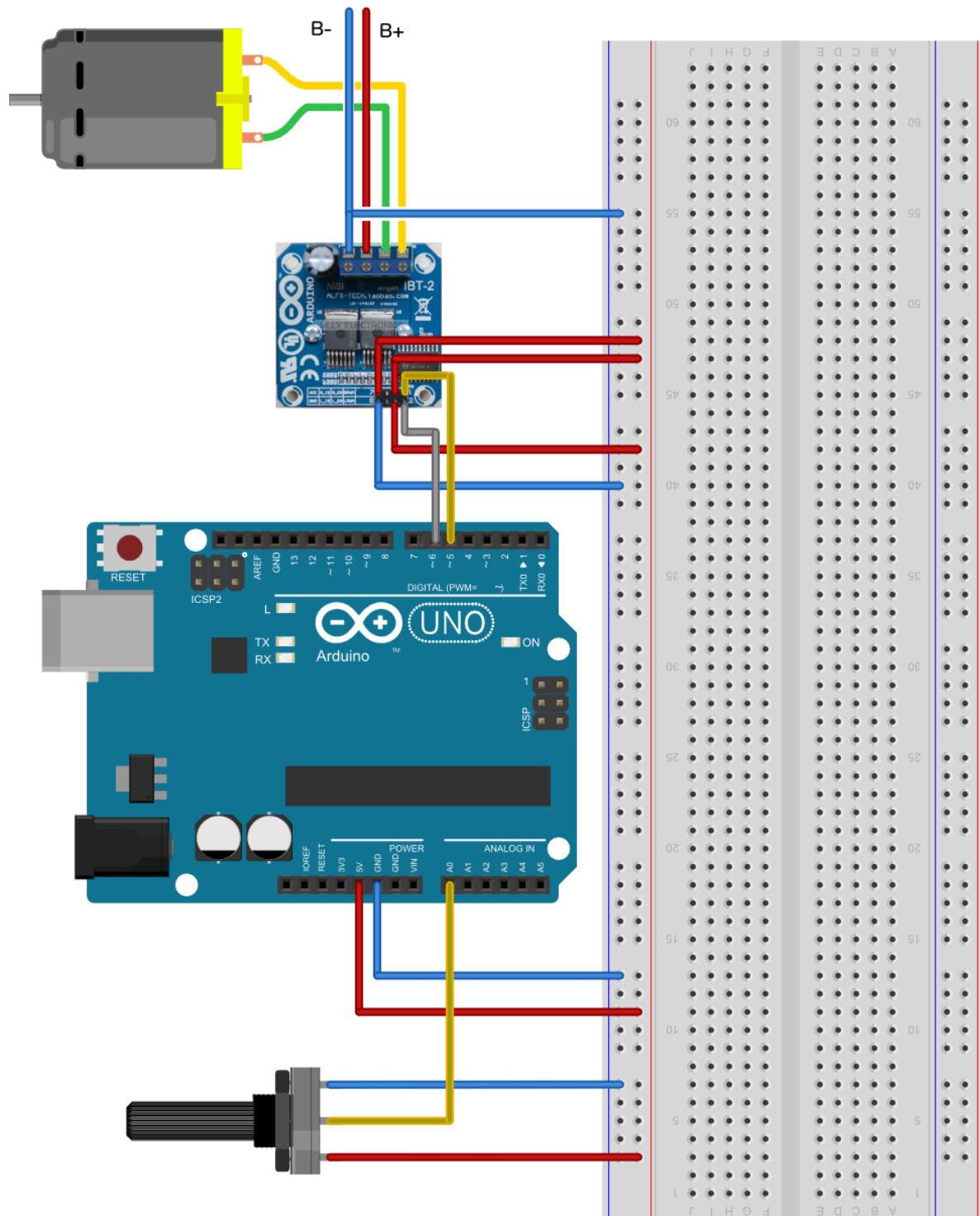
```

 9  IBT-2 pin 2 (LPWM) to Arduino pin 6(PWM)
10  IBT-2 pins 3 (R_EN), 4 (L_EN), 7 (VCC) to Arduino 5V pi
11  IBT-2 pin 8 (GND) to Arduino GND
12  IBT-2 pins 5 (R_IS) and 6 (L_IS) not connected
13  */
14
15  int SENSOR_PIN = 0; // center pin of the potentiometer
16
17  int RPWM_Output = 5; // Arduino PWM output pin 5; conne
18  int LPWM_Output = 6; // Arduino PWM output pin 6; conne
19
20  void setup()
21  {
22      pinMode(RPWM_Output, OUTPUT);
23      pinMode(LPWM_Output, OUTPUT);
24  }
25
26  void loop()
27  {
28      int sensorValue = analogRead(SENSOR_PIN);
29
30      // sensor value is in the range 0 to 1023
31      // the lower half of it we use for reverse rotation;
32      if (sensorValue < 512)
33      {
34          // reverse rotation
35          int reversePWM = -(sensorValue - 511) / 2;
36          analogWrite(LPWM_Output, 0);
37          analogWrite(RPWM_Output, reversePWM);
38      }
39      else
40      {
41          // forward rotation
42          int forwardPWM = (sensorValue - 512) / 2;
43          analogWrite(RPWM_Output, 0);
44          analogWrite(LPWM_Output, forwardPWM);
45      }
46  }

```

The following Fritzing diagram illustrates the wiring. B+ and B- at the top of the diagram represent the power supply for the motor. A 5k or 10k potentiometer is used to control the speed.





Dr Rainer Hessmer / December 28, 2013 / Robotics / Arduino

112 thoughts on “IBT-2 H-Bridge with Arduino”

**Vinicius** 👤

February 21, 2014 at 2:21 pm

Hi, do you know if it is possible to control 3 motors with only one Arduino Uno board? Thanks!

**Dave Corr**

November 16, 2020 at 5:54 pm

yes I have 3 IBT-2 controlled by arduino on my SIM RIG..

I have 3 of them connected.

Take a look

<https://www.youtube.com/watch?v=NwYMNQeXzXQ>

**Gareth Brian McGuire**

July 3, 2023 at 1:20 am

Hi Dave, I'm attempting to make racing simulators to sell here in South Africa.

Would you be interested giving me a boost financially? My email is

guypmb@gmail.com.

**mohamed**

October 21, 2022 at 5:09 am

no

**jpWA** 👤

March 2, 2014 at 4:03 pm

each h-bridge needs 2 pwm ports, and an arduino uno has 6, so yes

**Dmitri** 👤

June 28, 2014 at 12:04 am

Thank you very much, Dr. Hessmer!

**Benny** 👤

July 13, 2014 at 6:35 pm

Is there any reason why the two sides require a dedicated PWM pin? Could you tie those together to the one PWM output from the Arduino and use R_EN and L_EN to select direction? Even better, if one is always the inversion of the other, could a NOT gate be used so that to drive a motor forward and reverse, only one PWM pin and one non-PWM pin are required? (Possibly meaning as many as 6 motors could be driven from the one Arduino)

... Actually, writing that gives me an inkling as to why two PWM inputs are available...

Similarly, could two of these bridges be used to drive a high current stepper motor?

**David**

January 20, 2020 at 7:18 am

These boards definitely can't be used with a single PWM pin. The half-wit chinese board torturer who designed this thing named the enable pins R and L but there is no relation to the driving direction. They are just the enable pins for each of the two half-bridges on the board. (The L pin for the right of them and vice versa) Unless you activate them both, there is no function at all.

Therefore, it is not possible using only one PWM pin without further parts. With some additional logic you could easily use one PWM pin and one direction select pin.

I can't even articulate my anger against such people. This board must have consumed thousands of hours of good people trying to figure out why dhe

directional enable pins do not enable any direction, bringing despair to beginners and spoiling a nice hobby for so many.

Best Regards and good luck.



bruceg

June 9, 2022 at 9:24 pm

Arrrh, David,

I know this is 2 years late, but dis-information needs to be corrected whenever it is found.

The board is advertized as a dual half-bridge driver. If the user chooses to combine the 2 into a full-bridge that's fine. Invert the Left & Right PWM (in software or hardware) and you get bi-directional control of 1 brushed DC motor.

It also can drive a high power, 4 wire stepper motor (or more likely a pair of them at opposite sides of a heavy CNC work table) in reversable, single step mode. (Requires a special lookup table, but microcontrollers are quite good at reading these)

Also lots of DC motor applications don't need a reverse, just a high voltage, large power, High OR Low side PWM. This very versatile module can drive (up to 24V, 30A each) 2 uni-directional motors where the switching side for each is selected via setting Left (or Right) to high for high side driving or to low for low side sinking, then applying independant PWM to each Inhibit pin.

This board was thoughtfully designed for HOBBYISTS to be able to experiment with and learn from as well as being a robust (and extremely cheap) dual half bridge driver in finished projects.

bye.



Pillarama

July 22, 2023 at 1:11 am

Hi Dave, bruceg and anyone else reading. Sorry I'm even later to the party – it's now mid 2023 a full decade since this article was written! :P, but I 100% agree with bruceg – and since I found this today as one of the top results when searching Google for how to use this board, clearly contribution is still relevant in 2023.... If you're short on time / impatient – skip to the [Findings] paragraph – but I think it's important to understand how this board/chip works, so here's my long story of a contribution. Hopefully it will help someone – regardless, I sure learned a lot today! 😊

Firstly – I'm a long time hobby electronics enthusiast – I'm not a qualified electrical engineer or anything similar. Everything I say here is “to the best of my knowledge” and “based on the research I have done today” but please correct me on anything if you know better! :). I read through all the info and many of the very many comments on this page (and many other pages I found) and felt like I still don't understand this board well enough still. Unfortunately, I apparently didn't read enough of the comments until most of this was written, as later I found much of the info below already mentioned by others... I only had a vague understanding of H-Bridges from some YouTube I watched a few years ago. So I broke out my DMM, my scope and dug up the datasheets for the chips on the board and spent much of the day messing around with it and studying the datasheets. The results were interesting and satisfying. When you understand it, this is a useful and flexible board.

I think the labelling on this board, while arguable not necessarily wrong, is “subjective” and leads people to make wrong assumptions about how to use it. Things may become more clear when you trace the circuit and read the datasheet to understand how the chip works.

Addressing a few concerns/comments in some other comments here:

[R_EN and L_EN] – these are not “Direction Enable” pins they're connected to the “INH” or “Inhibit” pins on each chip. They actually control if the respective chip (Left or Right) is “active” or “sleeping”, When sleeping, that chip does not energise either of its two FETs in its half-bridge. So it absolutely makes sense to join these pins together and enable them simultaneously when using these two chips in a “full-bridge” configuration – which is what you use

for driving one motor in both directions. In a full-bridge configuration, you won't be able to energise the motor at all unless both half-bridges are enabled. Keep in mind that you should be able to drive two motors from this board (in one direction only!) by connecting each motor to only one of the two "M" pin and either B+ or B-. In this configuration you may want to keep the "L_EN" and "R_EN" pins separated.

[PWM] – I would say that these pins are not "stricly" PWM – they are actually connected to the "IN" pins on the half-bridge driver chips. What they actually do is decide if the "HIGH" or "LOW" side FET in each half-bridge is enabled. Of course neither is enabled unless the INH "Inhibit" pin is also HIGH. Driving these pins with PWM signals is really actually flipping the active of the two FETs in a half-bridge. Therefore if you flip them both at the same time with either the same or inverse logic – nothing will actually happen. This is why when you are driving a motor with two PWM pins you have to hold one of them steady (LOW or HIGH) and PWM pulse the other... Down below I'll explain why I think this is actually a waste of a PWM pin.. maybe you can already begin to see why...

[Cross conduction] A word on Cross conduction can ruin your day with some Full H-Bridges but I believe cross conduction is taken care of internally in the BTN79x0 package. It's a "Half-Bridge IC" not a bare pair of MOSFETS – and therefore I'm pretty sure you don't need to worry about creating shorts through cross Conduction. (heck if you did I'm sure I've have blown up my board many times over today!). Considering the IN pin (pin 2) determines whether the HIGH or LOW side of a given half bridge is active and that pin can only be HIGH or LOW, not both, you cannot influence whether both LOW and HIGH side mosfets are on at the same time. The potential could exist for Cross Conduction to occur when switching from HGH to LOW but the chip takes care of that with an internal dead time where neither FET are active when switching. See the datasheet section 5.4.2 "Dead Time Generation".

Keep in mind that cross conduction occurs when both mosfets in one single half-bridge are active at the same time. The same is not true when the both LOW side FETs (only) or both HIGH sides FETs (only) are active on both half-bridge. While it's very possible to do this, in a full-bridge configuration, all this does is create the same potential on both sides of the

driven motor and since the other FET in each Half-bridge cannot also be enabled this does completely nothing. 😊 Since each chip takes care of ensuring that both of its own HIGH and LOW sides cannot be active simultaneously if LOW sides of both chips are active, then both HIGH sides must therefore be inactive and no cross conduction occurs from B+ to B-.

(FINDINGS:)

Now as burceg suggests – yes it is definitely possible to drive the board with a single PWM pin by using a logic inverter such as Texas Instruments SN7404 Hex Inverter IC to create an additional “inverted” PWM signal – this is the “hardware” option he refers to. Simply connect one IN pin – these are the ones labelled “PWM” (eg Pin 1 on the boards 8 pin header) to a >NON< PWM pin of your microcontroller (yes non-PWM on the Micro to PWM on the Motor controller board – not a mistake), then connect that same pin to the first “A” (input) of the SN7404 – The pin called “1A” which is actually chip pin 1) and the output 1 of the chip – “1Y” which is Chip pin 2 to the other “IN” pin on your motor controller board – eg Pin 2 on the 8 pin input connector.

Now connect the PWM pin on your micro to BOTH L_EN and R_EN (eg pin 3 and 4).

Your Non-PWM pin now becomes a “direction” pin. When it's HIGH, one Half-Bridge will enable the HIGH side fet only and the other Half-Bridge will enable only the LOW side fet. This doesn't need to be PWM – it's only going to change as rapidly as you want the motor to change directions. Now you can apply PWM to the “Inhibit” pins (called L_EN and R_LEN) to adjust the speed at which those active FETs are turned on and off. This will cause both Half-Bridge ICs to go to sleep and wake up with the PWM signal. Unsure if this is an intentional/correct use of the Inhibit pin – but it does work and I don't see why not...

But in fact it's absolutely possible to drive this board with SINGLE PWM (plus a non PWM pin) and NO ADDITIONAL components. [Wait... I now see Jim already said something similar to this back in 2014!! I think my solution is slightly different, but in that case someone else maybe even said exactly this!]. Once you understand what's going on above you may realise that since the motor will only ever be energised when the two PWM signals are different – we can

in fact regard one as a "direction" pin (no PWM required) and the other as a "speed" pin (using PWM). The only catch here is that the logic state of the direction will also affect whether the motor is energised with the speed pin being HIGH or LOW. Therefore in software we must "invert" the speed PWM output for one of the directions, otherwise for one direction low PWM values will result in the motor running slow and high PWM values being fast but in the other direction, low PWM values will make the motor run fast and high PWM values will be slow. This is easy to do though with something like the following:

```
if (myDir==0) {  
  setMotor(mySpeed, myDir);  
} else {  
  setMotor(255-mySpeed, myDir);  
}  
  
void setMotor(int setSpeed, int setDirection) {  
...
```

I've tried both of the described methods above and both work fine as far as I can tell.



Casanave Daniel

September 12, 2024 at 11:52 am

Bonjour,

J'arrive... Je suis devant une problématique d'arduino méga qui doit commander les moteurs d'un suiveur solaire via 2 IBT2 Pas très doué, je ne pige pas la connectique.

Il est indiqué sur un plan de connecter aux pwm les bornes respectivement : 08; 09 et 10; 11 sur N1 et N2 des IBT2 ? Je voudrais que cela corresponde au 44 PWM 42 LPWM et 40RPWM. Je nage un peu... beaucoup.

Merci de me sauver Si vous pouvez m'aider. J'ai un schéma que je pourrais joindre.

D.Cas.

**Wolfhard Röhr**

July 20, 2025 at 3:05 am

Ja Du hast Recht. Habe versucht die Eingänge mit digitalen Ausgängen des Arduino zu verbinden und aus dem Programm anzusteuern. Hat nicht funktioniert. Für zwei DC Motore waren da 4 Arduino Ausgänge, die auch noch knapp waren, verbraucht. Leider war ich mir wegen der bescheuerten Beschriftung völlig sicher und habe gleich alles auf eine Leiterplatte gebaut. Eine dauerhafte Verbindung mit VCC 5V hätte gereicht

**Tom**

August 23, 2014 at 5:03 am

is it possible to ad a feedback potentiometer to the dc motor. and code in arduino so it will act like 180 degree servo.

Kind regards

Tom Jensen

**Sam**

January 5, 2023 at 9:46 am

Did you ever get this to work? I'm trying to make a radio controlled 2 axis gimbal with 24v motors at the moment and can't find anything similar using this IBT_2.

**Ilan Elron**

August 24, 2014 at 7:14 am

Dr. Hessmer,

I plan to use a similar (higher power) single H bridge to control a DC brushed motor. The bridge has a similar connection schema, including an ENABLE and LPWM and RPWM.

I am worried about cross-conduction (shoot-through) protection when changing direction. My guess is that the bridge does not provide it.

In that case, would such protection not be provided by always taking the xPWM pin that was fed until now to zero and ONLY THEN feeding the other?

In other words, in the code above, have the “forward rotation” coded thus:

```
// forward rotation
```

```
int forwardPWM = (sensorValue - 512) / 2;
```

```
analogWrite(RPWM_Output, 0);
```

```
analogWrite(LPWM_Output, forwardPWM);
```

Many thanks for your opinions and advice,

Ilan



Mike VP

October 23, 2017 at 5:19 am

Late reply, I know (better late than never in this case).

Yes this is an issue with these H-bridge models and high power uses.

The H-bridge does provide a delay (switch on/off delay / slew rate), which would prevent the cross-conduction issue if you could send both PWM signals at the same time. But because of the time arduino takes to process the line of code (analogWrite in this case) it will still cause cross-conduction.

I don't have the exact numbers anymore, but the build-in delay is about 4/6 ms (depending on resistance and temperature) and the delay time between the 2 lines of code is about 4-8 ms. This sometimes causes an overlap, resulting in crossconduction.

I broke about 2/3 h-bridges before I found out about this issue. The code you provides in your comment does solve the issue and no h-bridges have broken down since.

Someone also suggested using digitalWrite(pin, LOW), instead of analogWrite(pin, 0). But I don't know if this makes much difference.



YeNyi 👤

September 5, 2014 at 1:39 am

Hi Thanks for the post. Very useful.

May I know IBT-2 can control the Bipolar Stepper Motor, if so, how to connect the

motor.



Ray Edgley

December 12, 2018 at 12:44 pm

It is possible to drive a stepper motor with these bridges, however it is not recommended.

First you will need two as there are two sets of winding's in the stepper and this bridge will only handle 1 set of winding's

If you are using the IBT-2 I will assume you are planning to run a high power stepper.

In most cases, dedicated stepper drivers will not only control the voltage to the stepper motors, but also limit the current.

The voltage applied to the motor can be a lot higher than the nominal voltage of the winding's.

This is because at higher frequencies of phase reversal, the impedance of the winding's increases, reducing the current flow and motor torque will drop off rapidly. To overcome this, a higher voltage is used and the current is then limited to prevent damage to the motor.

This method allows motors that when stationary have a winding voltage of 6V to operate at up to 60V

So while it is possible, these motor drivers do not have the precise current control that would be needed but high power stepper motors.



Eman 👤

September 20, 2014 at 1:10 pm

Hi,

i want to ask why some time they connect R_EN and L_EN to a digital pin

thanks ^^



Ike 👤

October 6, 2014 at 7:58 am

Dr. Hessmer, thanks for sharing! Very informative and useful...I always learn something new and useful every time I visit your blog. Thx!

 **Chrs** 

October 14, 2014 at 11:59 pm

Good morning

This is exactly what I was looking for but would like to add rpm measurement via a slotted optical sensor board based on an LM393 chip and also an LCD display showing the measured rpm. Would you be able to provide some guidance on how to wire and program this additional equipment?

Thanks a lot

Chris

 **Jhovany**

April 22, 2016 at 1:36 am

LA Q TAL SI ES POSIBLE VER TUSA RPM HE IR CONTROLANDOLAS ASTA INALAMBRICAMENTE X BLUETOOTH PERO RECUERDA Q ESTE MODULO SOLO ES DE POTENCIA NESESITAS TU ENCODER Q TE DARA TUS RPM AHI ENCODER YA COMERCIALES MUY FACIL DE USAR EFECTIVAMENTE SON SENSORES OPTICOS LO Q TE PUEDO RECOMERDAR ES Q TRATES DE UTILIZAR AL IGUAL UN MOTOR CON ENCODER YA INCLUIDO Y UN LCD CON SU PUERTO I2C Y LISTO.

 **Jim K** 

November 29, 2014 at 5:03 pm

Thanks for the great info!! I thought I would share how I controlled this with one PWM pin. I “drove” both the enables with one PWM pin (Pin 44 goes to R_EN and L_EN). I then used the LPWM and RPWM pins to set the direction. I also use some digital pins to power/ground the logic of the IBT-2’s (46,47,52, 53).

The following code is for the Mega-2560 and can control two IBT-2's

```
int pwmpin1=44; // 8-bit pwm 0-255 timer 5 for motor 1
int pwmpin2=45; // 8-bit pwm 0-255 for motor 2
int mcgnd1=46; // ground for motor control 1
int mcpwr1=47; // power for motor control 1
int pwmdirection2r=48; // digital motor control direction 2
int pwmdirection2l=49; // digital motor control direction 2
int pwmdirection1r=50; // digital motor control direction 1
int pwmdirection1l=51; // digital motor control direction 1
int mcgnd2=52; // ground for motor control 2
int mcpwr2=53; // power for motor control 2

//Code snippet for two motors on 2 IBT-2's using just two pwm pins.

//set h-bridge to reverse or CCW
digitalWrite(pwmdirection1r,HIGH);
digitalWrite(pwmdirection1l,LOW);
analogWrite(pwmpin1,128);

delay(5000);

//else CW
digitalWrite(pwmdirection1r,LOW);
digitalWrite(pwmdirection1l,HIGH);
analogWrite(pwmpin1,128);

delay(5000);
//set h-bridge to reverse or CCW motor 2
digitalWrite(pwmdirection2r,HIGH);
digitalWrite(pwmdirection2l,LOW);
analogWrite(pwmpin2,128);

delay(5000);

//else CW
digitalWrite(pwmdirection2r,LOW);
```

```
digitalWrite(pwmdirection2l,HIGH);  
analogWrite(pwmpin2,128);
```

Jim K says:

November 30, 2014 at 4:51 pm

Could you share the fritzing part for the IBT-2?

Thanks,

Jim



murtadha

June 12, 2018 at 11:37 am

Hi can you send me this code please on my email

murtadhaalcount@gmail.com



fatima

September 17, 2020 at 11:11 am

hello, thanks for providing this code..

but the question is how did you use 4 pins to determine the direction of the two motors on this driver? or did you use relays or what?

can you send the connections please!

khwailehf@gmail.com



Omolo

October 8, 2020 at 8:22 am

Kindly Jim send me this code too haroldmondi@yahoo.com



Tom Jensen 

February 10, 2015 at 2:48 am

Hej dr.Hessmer do you now if its possible to run a segway on two ibt2 motor controller instead

Of the sabertooth motor controller with arduino.i am a newbie to arduino. if you can run it on

Ibt2 do you then got some exsampel code for that.

Kind Regards

Tom jensen



honest 👤

March 12, 2015 at 2:17 am

hey guys im looking for a arduino code to control a dc motor using an analogue stick controller



Juan 👤

April 27, 2015 at 1:33 pm

Hola.

Estoy buscando un código de Arduino para controlar un motor de corriente continua utilizando un puente, H 43 A BTS 7960B,



Juan 👤

May 4, 2015 at 1:03 pm

Hola,

Estoy intentando programar,ARDUINO UNO,con el puente H 43A BTS 7960 B, y no consigo que funcione el motor,estoy, con el código, IBT-2, el código me lo acepta bien pero no funciona el motor,agradecería que alguien me pasara un código.

Saludos..Juan....



LINTON BROWN

August 27, 2015 at 10:36 pm

HELLO SIR:

I CANNOT GET THIS PROGRAM TO VERIFY IN LINE 32.

if (sensorValue < 512)

I CAN GET THIS PROGRAM TO VERIFY IN LINE 32 WHEN I CHANGE IT TO THIS.

if (sensorValue = 512)

WHAT IS THE CORRECT LINE 32 STATEMENT.

I WILL BE LOOKING FOR YOUR REPLY.

THANK YOU

LINTON BROWN



LINTON BROWN

August 27, 2015 at 10:52 pm

HELLO AGAIN:

PLEASE LOOK AT LINE 32 ABOVE .WHEN I EMAILED YOU AND I HIT THE
SUBMIT COMMENT AND MY EMAIL QUESTION CHANGED FROM < to <.PLEASE
HELP ME

ON THIS. I HOPE YOU CAN UNDERSTAND ME.

LINTON BROWN



Mohamed Ihab

September 3, 2015 at 5:58 am

Can you please upload the fritzing file component for the IBT-2 driver ?

Thank you Dr. Hessmer

Mohamed Ehab



Dr Rainer Hessmer 

September 3, 2015 at 5:57 pm

Sorry to disappoint. I cheated when I created the Fritzing diagram and just
overlayed the wires on an image of the IBT-2.

**Omolo**

October 8, 2020 at 8:35 am

Hallo could someone help me with sketch which can run two IBT_2 connected to arduino R3

**Donald**

September 5, 2015 at 9:26 am

Hi Dr. Rainer,

I used these motor drivers and your recommendation to run a golf cart.

After nine holes, the cart reduced in power. It seems the battery running out of juice. Yet when I measured the voltage of the battery, it is at 13v. And the heat sink is not even warm.

Yet, when I get home, it was running fine again.

Can you offer some insight to solve this problem?

Thank you,
Donald

**Ryan**

September 18, 2015 at 4:23 pm

This chip won't switch higher than 40Hz. Just tried using it as the DC->AC component in a step up transformer and it can't handle signals over 40Hz and still deliver the power it claims to deliver.

**hassan**

October 30, 2015 at 6:45 am

I need the foot print of this driver i cant find its dimension



Swante

November 15, 2015 at 10:13 am

@hassan:

In this Zip-file you can also find the size of the IBT-2

<http://www.uctronics.com/download/U3537-BTS7960.zip>

@Hessmer

Are you sure about the B+ and B-?

According to the pdf and all images, the pin assignment is B-,B+,M+,M- (from left to right)

According to your drawing, -M,M+,B+,B-

Or does this depends on the version of the IBT-2?



Dr Rainer Hessmer 

May 31, 2016 at 9:42 pm

Sorry for the late response. You are right, the motor and battery connections in my diagram were wrong. I hope I did not cause any harm. I just uploaded a corrected image.



Matija

January 25, 2016 at 7:37 am

Hi.

I used the same drivers with electrical model cars – wondering if anyone has an idea how to get it in wifi version?

**agustin**

February 19, 2016 at 8:55 am

hi.

How much voltaje is the maximum for the VCC pin in the IBT-2 H-BRIDGE because i have been connecting 9V to that pin and the IBT2 stop working, I am not sure if I burn it or the arduino is the one damage.

**Amer**

March 17, 2016 at 2:56 am

hi

How can I get the Transfer Function of the driver (IBT-2 H-bridge module)? please.

**Ralph**

April 7, 2016 at 11:59 pm

@swante

the motor goes in next to the IBT-2 text

power source goes in the middle of the board and polarity matters

so the drawing above is wrong,

very funny when i was pwm'ing my power source

here is the right drawing

http://www.homofaciens.de/bilder/technik/rotary-encoder_010.htm**Daniel**

April 26, 2016 at 3:27 pm

Ralph, after badly connected power source to M+ and M- , your motor driver working?

**Ralph**

May 31, 2016 at 5:19 am

Hi Daniel,

i hooked up a wiper motor and it is still doing fine
so 10 to 15 amps is still fine after wrong connection

**Dr Rainer Hessmer** 

May 31, 2016 at 9:40 pm

My sincere apologies for messing up the motor and power connections. I
uploaded a corrected image. Ralph, thank you for the link to your diagram.

**Marketing Consultant**

December 28, 2018 at 5:10 am

I was reading your pages while on a boat. This stuff is great. This website is like
the cookies to my tasty soy milk. I just shared your article on Google Plus. When
I started my browser this page was running.

**Daniel**

April 26, 2016 at 11:41 am

I too connected 12V to M+ , M-. My driver not work. If I burn it driver?

**Turgay OLGAY**

May 3, 2016 at 12:45 pm

Dear Mr

Dr Rainer Hessmer

I want to make a sun tracker dual axis

- * I have arduino mega 2560 R3
- * I have 2 pcs BTS7960 (my motors are 24Vdc power supllly)
- * Analog sensors 4 photodiots array (4 serial for east-4 serial for west- 4 serial up -4 serial down)
- * 4 push button for manuel control (east-west-up-down)
- * 1 pcs wind speed sensor (output 0-5Vdc)
- * when sun set sistem will park in EAST

need your help for codes

Best regards

 **sean H**

June 2, 2017 at 11:00 pm

did you ever get this figured out, if so would you please share your code, I'm trying to do the same thing.

 **Oliver Schalles**

August 21, 2023 at 1:25 am

Hello Turgay –

I do use the same ibt controller for my dual axis parabol-suntracker –
It is not ready, until now – due to time' and financial' restrictions . But I am working on it...

Last testing was to implememt a IR remote control that shall drive the motors to its direction (die to them buttons of the remote) or stopping them...

I posted some years ago, already that project GiraSol on "Element14" ...

<https://community.element14.com/challenges-projects/project14/solarpowerdevices/b/blog/posts/project-girasol—oliver-schalles#comment-136373>

Maybe you might have a look on it ?

wishing you a nice day !

Oliver

**donnacarter0103**

July 6, 2016 at 4:35 am

Amazing and informative stuff. Is it work with 3D printer? I am working on an arduino 3D printer project.

**Emre**

November 8, 2016 at 10:49 am

Dear Dr Rainer Hessmer,

Can you please kindly help me for my project?

Especially I would like to drive peltier with this motor driver. I only need to control voltage of peltier from 0 to 12 volts. What should be the pins (input pins) wiring. I dont need any reverse current. I would like to only give some pwm signals from raspberry pi with python and control output voltage. Thanks for your kind help.

**sean H**

June 2, 2017 at 10:58 pm

did you ever get an answer on this, if so would you be willing to share you code?

**sean H**

June 2, 2017 at 10:59 pm

Nevermind, your using a raspberry pi lol

**Mat**

November 20, 2018 at 12:28 am

Did you ever figure this out? I can't get mine to work

**Christian**

December 10, 2016 at 10:49 am

Good morning Mr Hessmer

I have a strong doubt on the IBT-2, that is:

I have a 12V DC motor which absorbs a current of 30A in S1 service (max 38A in S2 service).

Can I safely use IBT-2 to operate the motor 30A?

Someone said that IBT-2 is small for this use.

I would use Arduino + potentiometer for feedback control.

Thanks in advance and waiting for your answer

Christian

**andrew wanstall**

December 12, 2016 at 2:16 am

Hi

Im using the Motor Driver BTS7960 43A for a 24v dc motor 200watt, im using a raspberry pi to control the pwm. When I first set up I was using a frequency of 100htz, but when running the motor and motor controller got extremely hot!. So I change the frequency to 800htz. Its now running cool and smooth but could anyone tell me if this a good setting please?.

Many thanks

Andy

**Christian**

December 13, 2016 at 8:15 am

Yes, the driver can get up to 25Khz.

Your Motor how much current nominally (A)?

Pingback: [Let's Get Moving! | javasjetzt](#)



Tim Kelly

February 10, 2017 at 6:17 am

Thanks ! Can you give a clue on using the IS pin for current limiting/monitoring ?

I don't understand the datasheet for that pin.

Thanks in advance



Leonardo Bisaro

April 21, 2017 at 5:06 pm

Hi everybody!

Thanks in advance for your attention

I think i connected all OK, and burn the arduino UNO with the described code.

But, I can't get to move the motor.

I have any doubt aboout the GND of 12 battery and the GND of arduino.

Can I connect all GND?

Thanks in advance!!

Leo



Didier Scheffers

April 26, 2017 at 8:26 am

Dear Dr Hessmer ,

Thanks for your effort . IBT-2 was working in minutes and your explanation made things very easy to understand....

 **Joe**

May 14, 2017 at 8:39 pm

Odd question, Is there any way I can get your fritzing part for the bts7960?

 **Jahmoy Bruce**

June 4, 2017 at 6:39 am

How can I get the BTS6070 in the Fritzing software? I have been trying but I cant seem to find it.

 **Shah**

June 15, 2017 at 8:24 am

Dear Dr Hessmer,
I have try this program code and its works fine...but how to change the input from VR (sensor A0) to read signal from the RC receiver...so i can control from the rc controller...tq.

 **marc**

July 12, 2017 at 3:59 am

pourriez-vous me donner le code et le schema de branchement de deux ibt-2 avec arduino uno et commander par un joystick

 **Shreck**

October 23, 2017 at 11:25 pm

Hi Dr Hessmer

Is power supply need to be same voltage as DC motor? Is there any step down module. What I mean is I have a 18v DC motor and 24v power supply will that burn the motor?



Mike VP

October 24, 2017 at 12:24 am

There is no step down module. You could run the 18vdc motor on 24vdc for a while, but it's not recommended. I've ran a 12vdc motor on 24vdc for a while (on accident) and it did not break down. Granted, it never recieved any pwm signal higher then 120 or so, but that's still 24v peaks going trough it.

These documents are also usefull if you want to take a closer look at how the h-bridge works:

<https://www.elecrow.com/download/IBT-2%20Schematic.pdf>

http://www.robotpower.com/downloads/BTS7960_v1.1_2004-12-07.pdf

<http://pdf1.alldatasheet.com/datasheet-pdf/view/15573/PHILIPS/74HC244D.html>



Jesse Grillo

December 26, 2018 at 10:05 pm

I signed up for your mailing list last week and I was wondering how often I'll be receiving Emails from you. I really enjoyed your page. I amofficially impressed, I need to say. I have just learned new insights because of your article. Funny thing I thought I was a pro before reading your site but it turns out I am a dummy. This is helpful to everyone who reads it.



Esau Montoya

November 27, 2017 at 7:56 pm

Hello Dr. Hessmer

Could you please tell me how to get the IBT-2 part to add it to a fritzing diagram?

Regards



January 29, 2018 at 8:01 am

Hii I m Smith Mia, If you need guidance for and Roku issues then we are here to serve best and optima lsolution to solve your issues.



March 15, 2018 at 12:38 pm

can i get fritzing part of ibt-2 driver?



April 20, 2018 at 7:21 am

I need to control a 12v DC motor using a BTS7960 motor driver on the basis of a digital input.

such as the motor have to run clock wise when a digital pin is high and when that digital pin is low , the motor should run counter clock wise.

what would be the code for performing this operation...?



April 27, 2018 at 10:27 am

Hi,

I can't find on the web how to use side current alarm pins R_IS and L_IS.

Is someone able to help me ?

Thanks



April 30, 2018 at 10:41 pm

I get an error at line 27 when I Verify or Upload the code above as follows:
“a function-definition is not allowed here before ‘{‘ token”
(at the line after void loop)
I am using Arduino IDE 1.8.5 with Windows 10
Can anyone advise the problem, & how to fix this please?
Thanks



Peter

June 14, 2018 at 10:20 am

Is it possible for a ibt-2 to control a motor set up as a servo as I wish to make a large high torque servo.
If I run the servo motor out put in to pin one and two will it fry?
Thanks Pete

<https://m.youtube.com/watch?v=aB9YIA-z0Js>



Kumara Fernando

July 5, 2018 at 4:15 am

my arduino bts7960 connect with ibt-2 and hall sensor .problem i having with out moving hall sensor motor start working slowly please hellp me

Thanks
Kumara Fernando



Kumara Fernando

July 5, 2018 at 4:15 am

my arduino bts7960 connect with ibt-2 and hall sensor .problem i having with out moving hall sensor motor start working slowly please hellp me no im not

Thanks

Kumara Fernando



Vahid

July 8, 2018 at 10:13 am

hi...i connected this drive to a 12 v battery suddenly the battery burned with fire and smoke!!!!why this thing happened?i should say that the was no consumer attached to drive...no motors no actuator...dose reverse polarity connection of battery cause this?B+ of battery to B- of driver..and B- of battery to B+ of driver.is my driver burn down too?



bruceg

June 9, 2022 at 9:43 pm

The mosfets used have ‘intrinsic’ built in reverse diodes. This just means that the fabrication of each transistor’s parts accidentally results in a diode between the drain part and the source part. This is good fo handling the ‘fly back’ pulse from an inductor (like an electric motor), BUT ...

If you connect a battery backwards it gets an almost dead short ($\sim 0.02R$) through the diodes which can handle $\sim 100A$ through a 2 volt drop = 200W – expect bad things to happen to the battery and/or mosfets.

bye.



property valuation fees

October 14, 2018 at 3:48 pm

It’s amazing to go to see this site and reading the views of all colleagues regarding this piece of writing, while I aam also zealous of getting experience.

Also visit mmy web page [property valuation fees](#)



October 17, 2018 at 3:47 am

Hi, Dr. Hessmer

May I ask for the ibt-2-h-bridge library for fritzing?

Sorry for my bad english

Thank you



November 7, 2018 at 9:56 am

Hi, Dr. Hessmer

May I ask for the ibt-2-h-bridge library for fritzing?

Tancks in advance

Pingback: [Farmaid: Robot Deteksi Penyakit Tanaman – Utak Atik ESP32/8266](#)



December 14, 2018 at 8:02 am

Hello.

Will this driver work directly with the RC receiver

FS-iA10B.

Regards.



January 9, 2019 at 2:43 pm

This product is much useful for drive motor. I just want to one thing that is Can I use this drive with JOYSTICK? If yes can you send code for that. Now I am using Arduino uno microcontroller. Please help me.

 **justine barcelo**

February 20, 2019 at 11:01 pm

hello, need to work with 2 dc motors, already done with wiring by following the posted diagram above but when I try to turn on (forward or reverse) motor 1, motor 2 GND pin is heating causing wire melt. how can this possibly be fix? please answer
Thanks

 **Giovanni greco**

March 31, 2019 at 9:26 am

vorrei usare Using Motor Driver BTS7960 per sostituire una board L298N che è limitata in corrente e si blocca!!

Nel mio esempio in arduino pilota il motorino dc con due interruttori NA per le due direzioni e un potenziometro per vel.

Tuttavia uso solo tre pin di Arduino due digitali IN ad uno per il PWM!

Ho appena comprato la nuova scheda ho cercato su internet ma non trovo risposte, come posso usare il Motor Driver BTS7960 per sostituire il L298N ?

lo schetch è ottimizzato per i tre pin....

! grazie!

 **Roberto**

April 4, 2019 at 10:56 am

It wanted Flickr to work seamlessly with yahoo answer app ([Roberto](#)) Mail.

 **li**

May 12, 2019 at 11:39 pm

what is the right board?I did not have it.I will change the IBM_2 board.
I only have a IBM-2 and a arduino.



James McWhinney

July 29, 2019 at 9:38 pm

I believe there is a typo in the image above for “Usage two:”
I think what they are trying to say is:

Usage 1:

VCC to 5V power

GND to arduino ground

R_EN and L_EN shorted to 5V

L_PWM connected to PWM signal for forward speed

R_PWM connected to PWM signal for reverse speed

Usage 2:

VCC to 5V power

GND to arduino ground

L_EN to digital arduino pin for forward

R_EN to digital arduino pin for reverse

L_PWM and R_PWM shorted to PWM signal for forward or reverse speed

(Your choice would depend on if you wanted to use up 2 PWM pins + 1 digital pin, or
2 digital pins and 1 PWM pin on your arduino)



David

January 20, 2020 at 7:26 am

Dear Mr. McWhinney,

Much to my regret, I must inform you, usage 2 is not possible. The two enable-pins do not correspond to directions but to the two half bridge ICs (And then R and L are swapped). Unless both are activated, it won't do anything. (Well, except changing the Voltage of the output to B- or to B+but there is no current flowing)



January 14, 2020 at 8:16 am

hi my question is can i control two motors using single IBT-2 driver? i am making a wheel chair ,in order to rotate i must stop 1 and run the other one .any info about that??
thanks



January 24, 2020 at 2:24 am

Hi, I followed your instructions to use IBT-2 to control a motor, used the same arduino sketch and circuit. 12V battery connected to -B&+B, arduino UNO/Nano VCC reads 4.75V (Chinese clone) connected to VCC and GND, but not success,

Tested with an multimeter and found the the below
When sensorValue 0V to 10V.

but (-B & +M)and (-B & -M) reads >0 to 10V, (changes according to pot value)

and (+B & +M) or (+B & -M) reads 0v.

I assume that +10V comes in both +M & -M.

While sensorValue>512 voltage across
+M & -M is 0V only, suppose to be >0 to 10v.

but +B & -M or +B & -M reads >0 to 10V,
I assume that both +M and -M is now negative or 0v.

And further

I used some simple sketch to make L_EN and R_EN HIGH and RPWM HIGH and LPWM LOW. The voltage reads same as above. Please note that while using 12V battery BTS9600 gets too hot, hence I used a 7.5lipo.

Any advise please.

<http://www.hessmer.org/blog/2013/12/28/ibt-2-h-bridge-with-arduino/>



Jaleel

January 24, 2020 at 2:44 am

Hi, please delete my previous post as it mistyped some info,
I followed This instructions to use IBT-2 to control a motor, used the same arduino sketch and circuit. 12V battery connected to -B&+B, arduino UNO/Nano VCC reads 4.75V (Chinese clone) connected to VCC and GND, but not success,

Tested with an multimeter and found the the below
When sensorValue 0V to 10V.

but (-B & +M)and (-B & -M) reads >0 to 10V, (changes according to pot value)

and (+B & +M) or (+B & -M) reads 0v.

I assume that +10V comes in both +M & -M.

While sensorValue > 512 voltage across
+M & -M is 0V only, suppose to be >0 to 10v.

but +B & -M or +B & -M reads >0 to 10V,

I assume that both +M and -M is now negative or 0v.

And further

I used some simple sketch to make L_EN and R_EN HIGH and RPWM HIGH and LPWM LOW. The voltage reads same as above.

Any advise please.



Mohammed Rizwan Uddin

March 1, 2020 at 8:39 am

we can use mpu6050 to control ibt-2



Carneal

March 10, 2020 at 12:29 pm

I'll subscribe to that writer and his future books. The post is well organized, reminded me about <https://usagrantsforsinglemothers.com/>. I see the author has a real knack for this topic. I like that theme, and I'm in constant search of new bits and the most recent news. I really enjoyed that one, because it is filled with interesting facts and it is a kind of easygoing article. I spent only a few minutes reading and due to well-structured text, I understand it completely. Thanks!



Joe

April 19, 2020 at 5:57 am

What are R_IS and L_IS used for?
And why aren't they connected to anything?



Thomas S Low

May 21, 2020 at 5:17 pm

Yes, it certainly IS possible to use TWO of these half bridges to drive the TWO coils of a BIPOLAR (4 wire) STEPPER MOTOR. You can use EXISTING Arduino code without ANY editing (from the site below), if you use the same 4 Arduino DIGITAL pins as inputs to the TWO half bridges.

See the excellent tutorial here:

<https://www.arduino.cc/en/Tutorial/StepperOneStepAtATime>

and specifically

<https://www.arduino.cc/en/Reference/StepperStep>

Wiring from the Arduino UNO to the TWO half bridge boards is as follows

NOTE: this is to connect to a single BIPOLAR STEPPER MOTOR Bipolar (4 wire, 2 coils)

Polarity of connection the coils doesn't matter, so long as ONE coil is connected to M- and M+ on ONE half bridge output terminal, and the OTHER coil is connected M- and M+ on the OTHER half bridge output terminal. Inverting the polarity on either coil simply changes the motor rotation direction for a given stepper pulse. (and the stepper motor can be stepped in EITHER direction by changing the sign of the number of steps in the Arduino sketch.)

suffixes _1, _2 indicate which of TWO half bridge boards (and which stepper coil) is involved

CONNECTIONS to FIRST IBT-2 H-bridge module half bridge:

VCC_1 to arduino +5V power (from Arduino UNO or similar)

GND_1 to arduino ground

L_EN_1 to arduino +5V power (ENABLE= TRUE)

R_EN_1 to arduino +5V power (ENABLE= TRUE)

L_PWM_1 = arduino digital pin 8

R_PWM_1 = arduino digital pin 9

STEPPER MOTOR CONNECTION (COIL_1)

B-_1 = GND of STEPPER MOTOR SUPPLY

B+_1 = Vm of STEPPER MOTOR SUPPLY (note Vm can be DIFFERENT from Arduino Vcc

and supply needs enough current for motor (1A or more))

M-_1 = one end of STEPPER MOTOR COIL_1

M+_1 = other end of STEPPER MOTOR COIL_1 (invert M- & M+ still works but direction reverses)

CONNECTIONS to SECOND IBT-2 H-bridge module half bridge:

VCC_2 to arduino +5V power (from Arduino UNO or similar)

GND_2 to arduino ground

L_EN_2 to arduino +5V power (ENABLE= TRUE)

R_EN_2 to arduino +5V power (ENABLE= TRUE)

L_PWM_2 = arduino digital pin 10

R_PWM_2 = arduino digital pin 11

STEPPER MOTOR CONNECTIONS

B-_2 = GND of STEPPER MOTOR SUPPLY

B+_2 = Vm of STEPPER MOTOR SUPPLY (note Vm can be DIFFERENT from Arduino Vcc

and supply needs enough current for motor (1A or more))

M-_2 = one end of STEPPER MOTOR COIL_2

M+_2 = other end of STEPPER MOTOR COIL_2 (invert M- & M+ still works but direction reverses)

NOTE: you can provide a suitably LOW voltage and HIGH current power supply to run the stepper motor (connected to B- and B+ on the half bridge output terminal). This supply is ****separate**** from the small +5V supply coming from the Arduino into each Vcc pin on each half bridge. You may need to insert a suitable high power low Ohm value resistor Rs (e.g., say a 1 Ohm 5 Watt resistor), in series with the stepper motor supply to limit the current, depending on the stepper motor coil resistance (and inductance, if you are running it fast).

In my case, I was able to omit the Rs current limit, and simply use a direct connection of my particular motor to a ****separate**** high current Vm supply (which coincidentally ALSO was at +5V, but could have been chosen otherwise, depending on motor specs and torque requirements.)

Pingback: [IBT-2 H-Bridge with Arduino – Dr Rainer Hessmer | Podex' Tagebuch](#)

 **Fabrizio Esposito**

January 31, 2021 at 2:24 am

Dear Dr Hessmer ,

Thanks for your effort and support. I am working on the design of a robot with 2 24V motors. I would like control motor speed remotely, using NODEMCU and smartphone BLYNK application. If I wire DC Motors with L298N driver, they works

better but they stop when I give full PWM. If I replace L298N with IBT-2 its worst.
Could you give me some help?

 **Tomek**

July 12, 2021 at 1:34 pm

Hi

Did you manage to get the controls on the IBT-2?

 **Ahmad Hamza**

September 1, 2021 at 12:48 am

code for motor Left , Right , reverse and stop using push botton for IBT2 driver

 **Ahmad Hamza**

September 1, 2021 at 12:53 am

code for motor Left , Right , reverse and stop using push botton for IBT2 driver
I can't understand how I can do my wheel chair right,left,stop and reverse It is
going forward only.
Thanks

 **Ahmad Hamza**

September 1, 2021 at 12:54 am

Code for motor Left , Right , reverse and stop using push botton for IBT2 driver
I can't understand how I can do my wheel chair right,left,stop and reverse It is going
forward only.
Thanks

 **zubair**

May 31, 2022 at 12:52 pm

i want to drive 12volte 6 amp motor without potentiometer meter plz give me code thanks.



Karsten

October 10, 2022 at 2:50 pm

Hello,

i am using an adruino and an IBT-2 for controlling ham radio antennas for satellite comunication. I have one issue, I would like to implement a hardware end switch for the motors. If i switch off EN_L or EN_R i am not able to run in the opposite direction. Do I have to switch off the PWM signal to get a hardware switch for one direction and I am then able to run the other direction?!

Best regards,

karsten



Conrad Weston

November 4, 2022 at 6:27 pm

I found this all very interesting and extremely helpful. All of my current questions have been answered somewhere in this list of comments. I may have more in the future, but for now, thank you.

Regards,

Conrad R. Weston.



Stephan Abonyi

May 2, 2023 at 8:51 am

Ich habe eim Attiny85 eingesetzt mit 8kz pwm. In das Austo eingebaut und nach 4-5 Stunden war der eine IC defekt geworden. Der KühllkOrper wurde mit der Hauptgebse gekühlt-trotz dem.

(Der Gebäsemotor ziht 8A bei voller Leistung) Schade!

**Ingrid Wordsworth**

July 5, 2025 at 1:41 am

After reading this article, I have a new understanding of this topic. Thank you very much for the author's careful organization and sharing.

Dr Rainer Hessmer / Proudly powered by WordPress